THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 3: **Pair of Linear Equations in Two Variables** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The solution of $5x + 2y = 20$, $3x + 4y = 26$ has $y =$ 1

(a) -5 **(b)** 7 **(c)** 2 **(d)** 5

2. For the pair $x + y = 20$ and $x - y = 4$, the value of xy is: 1

(a) 192 **(b)** 20 **(c)** 96 **(d)** 104

3. The pair $x + 4y = 3$ and $2x + ky = 7$ has no solution when $k =$ 1

(a) 9 **(b)** 6 **(c)** 8 **(d)** 4

4. The point where the lines $9x + 2y = 50$ and $5x + 3y = 41$ intersect has $x + y =$ 1

(a) 11 **(b)** 7 **(c)** -3 **(d)** 4

5. The pair of equations $5x + 4y = 3$ and $10x + 8y = 6$ is: 1

(a) exactly two solutions
(b) infinitely many solutions (coincident lines)
(c) no solution (parallel lines)
(d) exactly one solution (intersecting lines)

6. The solution of $x + 3y = 12$, $2x + y = 14$ has $x =$ 1

(a) 2 **(b)** 8 **(c)** -6 **(d)** 6

7. The solution of $2x - 3y = 6$, $3x + 2y = 22$ has $y =$ 1

(a) 6 **(b)** 8 **(c)** -2 **(d)** 2

8. For the pair $x + y = 24$ and $x - y = 8$, the value of xy is: 1

(a) 160 **(b)** 24 **(c)** 256 **(d)** 128

9. The pair $x + 5y = 2$ and $3x + ky = 7$ has no solution when $k =$ 1

(a) 5 **(b)** 16 **(c)** 15 **(d)** 6

10. The point where the lines $8x + 3y = 30$ and $3x + 5y = 19$ intersect has $x + y =$ 1

(a) 3 **(b)** 2 **(c)** 1 **(d)** 5

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the value of k for which the pair $x + 2y = 5$, $4x + ky = 21$ has no solution. 2

12. A man tells his daughter, "Seven years ago I was 8 times as old as you were. Three years from now I shall be 3 times as old as you will be." Form the pair of linear equations (do not solve). 2

13. Solve by substitution: $3x + y = 11$, $x + y = 7$. 2

14. Find the value of k for which the pair $x + 4y = 3$, $2x + ky = 7$ has no solution. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. On comparing the ratios, find whether the lines $2x + 3y = 8$ and $4x + 6y = 20$ intersect, are parallel or coincide. 3

16. Solve by elimination method: $2x - 3y = 6$ and $3x + 2y = 22$. 3

17. The sum of the digits of a two-digit number is 11. If the digits are interchanged, the number becomes 63 less. Find the number. 3

OR

On comparing the ratios, find whether the lines $7x - 3y = 11$ and $14x - 6y = 22$ intersect, are parallel or coincide. 3

18. Solve by elimination method: $3x + 4y = 20$ and $5x - 2y = 16$. 3

OR

The sum of the digits of a two-digit number is 11. If the digits are interchanged, the number becomes 27 less. Find the number. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A fraction becomes $\frac{2}{3}$ when 1 is added to both the numerator and the denominator, and becomes $\frac{8}{11}$ when 3 is added to both. Find the fraction. 5

OR

A man travels 480 km partly by train and partly by bus. If he travels 240 km by train and the rest by bus, he takes 7 hours; if he travels 300 km by train and the rest by bus, he takes $\frac{29}{4}$ hours. Find the speeds of the train and the bus. 5

20. A fraction becomes $\frac{6}{11}$ when 2 is added to both the numerator and the denominator, and becomes $\frac{9}{14}$ when 5 is added to both. Find the fraction. 5

OR

A man travels 360 km partly by train and partly by bus. If he travels 160 km by train and the rest by bus, he takes 8 hours; if he travels 200 km by train and the rest by bus, he takes $\frac{41}{5}$ hours. Find the speeds of the train and the bus. 5

END OF QUESTION PAPER — **Total: 40 Marks**